SIMATS ENGINEERING
ASSESSMENT TOOL 3
MCQ Test

COURSE NAME: Computer Networks **COURSE CODE: CSA0711**

COURSE OUTCOME COVERED

CO3: *Analyze the performance of core network protocols such as TCP, UDP, HTTP, and DNS under varying conditions*

Assessment Tool Weightage: 20%

Code of Conduct:

I **K. Sampath Reddy (192424285)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: **K. Sampath Reddy**

QUESTIONS:20 **Total Marks:20**

1. Which TCP mechanism reduces the congestion window by half upon detecting packet loss via duplicate ACKs?
   a) Slow Start
   b) Fast Retransmit
   c) Fast Recovery
   d) Tahoe Reset
   (BL4)(1)

2. A TCP sender has CWND = 8 MSS and SSTHRESH = 16. After one RTT with no loss, CWND becomes:
   a) 9 MSS
   b) 16 MSS
   c) 12 MSS
   d) 8 MSS
   (BL4)(1)

3. Which field in the TCP header is used to implement flow control at the receiver side?
   a) Sequence number
   b) Acknowledgement number
   c) Window size
   d) Urgent pointer
   (BL4)(1)

4. Nagle's algorithm causes increased latency primarily because it:
   a) Disables ACK piggybacking
   b) Buffers small segments until ACK arrives
   c) Reduces CWND on every packet
   d) Increases SYN backlog
   (BL4)(1)

5. In a high-latency satellite link (600 ms RTT), TCP throughput is primarily limited by:
   a) MTU size
   b) CWND growth rate relative to RTT
   c) DNS resolution time
   d) UDP competition

(BL4)(1)

6. UDP is preferred over TCP for DNS queries primarily because:
   a) UDP supports fragmentation
   b) DNS responses are small and connection overhead is unnecessary
   c) TCP cannot use port 53
   d) UDP has better error correction
   (BL4)(1)

7. A UDP-based streaming application experiences 15% packet loss. The best application-layer mitigation is:
   a) Increase UDP buffer size
   b) Switch to TCP
   c) Implement Forward Error Correction (FEC)
   d) Reduce frame rate to 1 fps
   (BL4)(1)

8. What is the maximum UDP payload size without IP-level fragmentation on a standard Ethernet link (MTU = 1500 bytes)?
   a) 1480 bytes
   b) 1472 bytes
   c) 1460 bytes
   d) 1500 bytes
   (BL4)(1)

9. QUIC protocol uses UDP as its transport. The key advantage over TCP-based TLS is:
   a) Eliminating IP header overhead
   b) 0-RTT or 1-RTT connection setup
   c) Removing encryption
   d) Native multicast support
   (BL4)(1)

10. HTTP/1.1 persistent connections reduce latency by:
   a) Enabling UDP fallback
   b) Reusing the same TCP connection for multiple requests
   c) Compressing HTTP headers
   d) Sending requests without waiting for DNS
   (BL4)(1)

11. Head-of-line blocking in HTTP/1.1 occurs when:
a) The DNS server is slow
b) A response blocks subsequent responses on the same connection
c) TCP CWND drops to zero
d) The server rejects pipelined requests
(BL4)(1)

12. HTTP/2 multiplexing solves HOL blocking at the HTTP layer but not at the:
a) Application layer
b) DNS layer
c) TCP layer
d) IP layer
(BL4)(1)

13. Which HTTP status code indicates that a resource has permanently moved and browsers should update bookmarks?
a) 301
b) 302
c) 304
d) 307
(BL4)(1)

14. In HTTP caching, the Cache-Control: no-cache directive means:
a) Never cache this resource
b) Validate with the server before using a cached copy
c) Cache for 0 seconds
d) Disable all caching headers
(BL4)(1)

15. Which DNS record type maps a domain name to an IPv6 address?
a) A
b) CNAME
c) AAAA
d) MX
(BL4)(1)

16. A DNS resolver switches from UDP to TCP when:
a) The query type is MX

b) The response is truncated (TC bit set)
c) TTL exceeds 3600 s
d) The authoritative server is unreachable
(BL4)(1)

17. Which attack exploits the UDP-based DNS protocol by sending forged responses with a matching transaction ID?
a) ARP poisoning
b) DNS cache poisoning
c) SYN flooding
d) BGP hijacking
(BL4)(1)

18. Setting a very low TTL (e.g., 30 s) for a DNS record primarily increases:
a) Cache hit ratio
b) Recursive resolver query load
c) DNSSEC signature validity
d) Zone transfer frequency
(BL4)(1)

19. A user browses to a new website. The correct sequence of protocol interactions for the first byte of the page is:
a) HTTP → TCP → DNS
b) DNS → TCP handshake → HTTP GET
c) TCP → DNS → HTTP
d) DNS → HTTP → TCP
(BL4)(1)

20. Under 20% packet loss, which protocol combination gives the best performance for a real-time multiplayer game?
a) TCP for game state, HTTP for assets
b) UDP with application-layer sequencing for game state, HTTP/3 for assets
c) HTTP/2 for all traffic
d) TCP with Nagle enabled for all traffic
(BL4)(1)

RUBRICS FOR CALCULATION:

Criteria	**Weightage (%)**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Conceptual Understanding	30%	Demonstrates thorough understanding; answers all questions accurately	Shows good understanding with few errors	Partial understanding; several incorrect responses	Lacks understanding; majority of answers incorrect
Accuracy of Answers	25%	All answers are correct	Most answers are correct with minor mistakes	Some correct answers; frequent errors	Mostly incorrect answers
Application of Knowledge	20%	Correctly applies concepts to problem-based questions	Applies concepts with minor errors	Limited ability to apply concepts	Unable to apply concepts
Time Management	15%	Completes quiz well within time efficiently	Completes on time with minor delays	Requires extra time or rushes through questions	Unable to complete within given time
Question Interpretation	10%	Interprets all questions correctly and responds appropriately	Misinterprets a few questions	Often misinterprets questions	Unable to understand questions

ANSWER KEY WITH EXPLANATIONS

(CSA07 – CO3 – Assessment Tool 3 – MCQ Test)

1. Which TCP mechanism reduces the congestion window by half upon detecting packet loss via duplicate ACKs?

Correct Answer: c) Fast Recovery

Reason / Solution: On three duplicate ACKs, Fast Retransmit resends the lost segment immediately without waiting for a timeout, but it is Fast Recovery that actually halves the window: it sets SSTHRESH = CWND/2 and adjusts CWND accordingly before resuming linear growth, avoiding a full return to Slow Start.

2. A TCP sender has CWND = 8 MSS and SSTHRESH = 16. After one RTT with no loss, CWND becomes:

Correct Answer: b) 16 MSS

Reason / Solution: Since CWND (8) is still below SSTHRESH (16), the connection is in Slow Start. In Slow Start, CWND grows exponentially and roughly doubles every RTT when ACKs are received with no loss, so CWND becomes $8 \times 2 = 16$ MSS.

3. Which field in the TCP header is used to implement flow control at the receiver side?

Correct Answer: c) Window size

Reason / Solution: The 16-bit Window Size field advertises how much buffer space the receiver currently has available, allowing the sender to throttle its transmission rate and implement receiver-side flow control.

4. Nagle's algorithm causes increased latency primarily because it:

Correct Answer: b) Buffers small segments until ACK arrives

Reason / Solution: Nagle's algorithm delays sending small segments and coalesces them until an outstanding ACK is received, which reduces the number of small packets on the network but adds latency for interactive/small-write applications.

5. In a high-latency satellite link (600 ms RTT), TCP throughput is primarily limited by:

Correct Answer: b) CWND growth rate relative to RTT

Reason / Solution: Throughput is bounded by CWND/RTT. With a 600 ms RTT, CWND takes far longer (in real time) to grow to the bandwidth-delay product than on a low-latency link, so the slow, RTT-paced growth of CWND — not MTU or DNS — is the limiting factor.

6. UDP is preferred over TCP for DNS queries primarily because:

Correct Answer: b) DNS responses are small and connection overhead is unnecessary

Reason / Solution: Most DNS queries and responses fit in a single small UDP datagram, so the 3-way handshake and connection-management overhead of TCP would add unnecessary latency for a simple request/response exchange.

7. A UDP-based streaming application experiences 15% packet loss. The best application-layer mitigation is:

Correct Answer: c) Implement Forward Error Correction (FEC)

Reason / Solution: FEC adds redundant data so the receiver can reconstruct lost packets without retransmission, which suits real-time UDP streaming where retransmission (as in TCP) would add unacceptable delay; larger buffers or lower frame rate don't address the underlying loss.

8. What is the maximum UDP payload size without IP-level fragmentation on a standard Ethernet link (MTU = 1500 bytes)?

Correct Answer: b) 1472 bytes

Reason / Solution: With a 1500-byte Ethernet MTU, the IP header (20 bytes) and UDP header (8 bytes) are subtracted first: $1500 - 20 - 8 = 1472$ bytes of usable UDP payload before IP fragmentation occurs.

9. QUIC protocol uses UDP as its transport. The key advantage over TCP-based TLS is:

Correct Answer: b) 0-RTT or 1-RTT connection setup

Reason / Solution: QUIC integrates the transport and TLS 1.3 handshakes over UDP, allowing a new connection to start in one round trip (or zero, for resumed connections), which is faster than TCP's separate TCP handshake followed by a TLS handshake.

10. HTTP/1.1 persistent connections reduce latency by:

Correct Answer: b) Reusing the same TCP connection for multiple requests

Reason / Solution: Persistent connections avoid repeating the TCP (and TLS) handshake for every request, letting multiple HTTP requests/responses share one already-established connection, which cuts round trips and latency.

11. Head-of-line blocking in HTTP/1.1 occurs when:

Correct Answer: b) A response blocks subsequent responses on the same connection

Reason / Solution: HTTP/1.1 responses on a single connection must be returned in the order requests were sent, so a slow response at the head of the queue blocks all responses behind it, even if their data is already ready.

12. HTTP/2 multiplexing solves HOL blocking at the HTTP layer but not at the:

Correct Answer: c) TCP layer

Reason / Solution: HTTP/2 multiplexes independent streams over one TCP connection, removing HOL blocking at the HTTP level, but because TCP itself delivers bytes in strict sequence, a single lost TCP segment still stalls all streams until it is retransmitted — HOL blocking reappears at the transport layer.

13. Which HTTP status code indicates that a resource has permanently moved and browsers should update bookmarks?

Correct Answer: a) 301

Reason / Solution: HTTP 301 (Moved Permanently) tells the client the resource now has a new permanent URL, so browsers and crawlers should update bookmarks/links to the new location; 302/307 are temporary redirects and 304 means "Not Modified".

14. In HTTP caching, the Cache-Control: no-cache directive means:

Correct Answer: b) Validate with the server before using a cached copy

Reason / Solution: Cache-Control: no-cache does not forbid storing the response; it requires the cache to revalidate with the origin server (e.g., via ETag/If-Modified-Since) before serving a stored copy. Preventing storage entirely is the purpose of no-store.

15. Which DNS record type maps a domain name to an IPv6 address?

Correct Answer: c) AAAA

Reason / Solution: An AAAA record maps a hostname to a 128-bit IPv6 address, analogous to how an A record maps a hostname to a 32-bit IPv4 address.

16. A DNS resolver switches from UDP to TCP when:

Correct Answer: b) The response is truncated (TC bit set)

Reason / Solution: When a DNS response is too large to fit in a single UDP datagram, the server sets the TC (truncated) bit; the resolver then reissues the query over TCP, which supports larger, reliable, segmented responses.

17. Which attack exploits the UDP-based DNS protocol by sending forged responses with a matching transaction ID?

Correct Answer: b) DNS cache poisoning

Reason / Solution: DNS cache poisoning (DNS spoofing) exploits UDP's lack of connection state by racing forged responses with a guessed/matching transaction ID and source port to a resolver's query, tricking it into caching a malicious IP address.

18. Setting a very low TTL (e.g., 30 s) for a DNS record primarily increases:

Correct Answer: b) Recursive resolver query load

Reason / Solution: A very low TTL forces caches to expire and discard the record almost immediately, so recursive resolvers must re-query the authoritative server far more often, increasing query load (in exchange for records propagating/updating faster).

19. A user browses to a new website. The correct sequence of protocol interactions for the first byte of the page is:

Correct Answer: b) DNS → TCP handshake → HTTP GET

Reason / Solution: The browser must first resolve the domain name to an IP address via DNS, then establish a TCP connection (3-way handshake) to that IP, and only then send the HTTP GET request to retrieve the first byte of the page.

20. Under 20% packet loss, which protocol combination gives the best performance for a real-time multiplayer game?

Correct Answer: b) UDP with application-layer sequencing for game state, HTTP/3 for assets

Reason / Solution: Under significant loss, TCP's retransmission and head-of-line blocking would stall time-sensitive game state, so UDP with custom, application-level sequencing/interpolation tolerates loss gracefully for real-time state, while HTTP/3 (QUIC/UDP) still gives reliable, loss-resilient delivery for supporting assets.